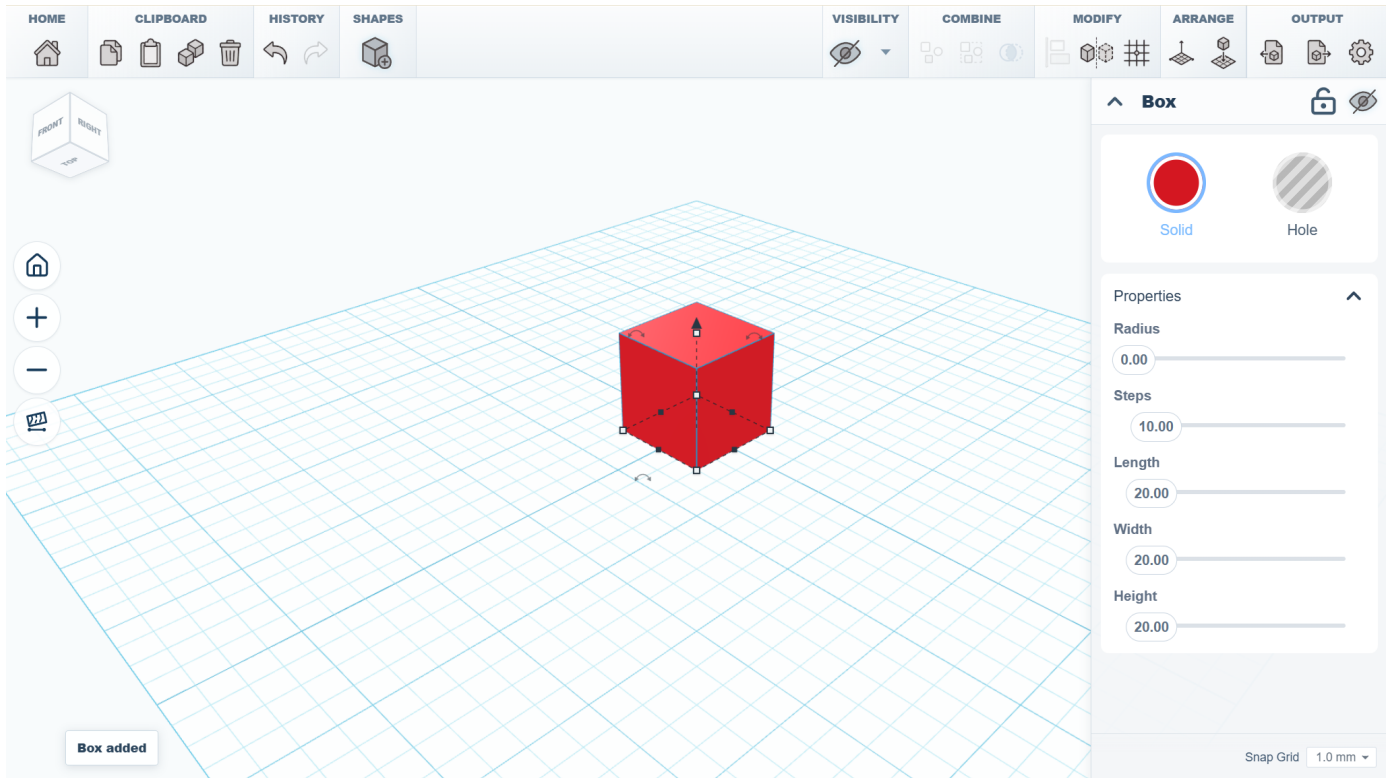


SKETCHFORGE

User Manual

Keyboard shortcuts, basic workflows, and complete tool reference

VERSION 0.9 / AUGUST 2026



Create precise models from primitives or custom sketches, prepare fabrication exports, and keep editable projects local to your browser.

Contents

Sections and major topics

1. Getting Started	5
What SketchForge Does	5
Browser Requirements	5
Create Your First Model	5
Dashboard Basics	5
2. Navigating The Editor	6
Main Areas	6
Mouse And Camera Controls	6
3. Creating And Editing 3D Geometry	7
Primitive Shape Library	7
Selection	7
Transform Handles	7
Shape Inspector	8
Moving With The Keyboard	8
Clipboard And History	8
Visibility, Locking, And Separation	8
4. Placement And Construction Planes	9
Placement Workplane	9
Persistent Construction Planes	9
Plane Workflow Tips	9
5. Combining And Modifying Geometry	10
Solids And Holes	10
Group And Ungroup	10
Boolean Intersection	10
Align	10
Mirror	10
Fillet And Chamfer	11
Boolean And Edge Limitations	11
6. Measurement Tools	12
Selection Dimensions	12
3D Ruler	12
Sketch Measure And Dimension	12
7. The 2D Sketch Workspace	13

Start Or Edit A Sketch	13
Sketch Navigation	13
Drawing Tools	14
Point And Curve Editing	14
Snapping	14
Constraints	15
Dimensions	15
Reference Images	15
Sketch Text	15
Project Geometry Into A Sketch	15
Offset	16
Sketch Mirror And Patterns	16
8. Turning Sketches Into 3D Models	17
Extrude	17
Revolve	17
Sweep	17
9. Import, Export, And Project Files	18
Import Formats	18
Export Formats	18
SKF Project Packages	18
Download Destinations	19
10. Workspace Settings	20
Appearance	20
Measurement	20
Workplane	20
History	20
Dashboard Export Settings	20
11. Complete Keyboard Shortcut Reference	21
Modifier Names	21
Project And Selection Shortcuts	21
Movement And Modeling Shortcuts	22
Viewport And Workplane Shortcuts	22
Pointer Modifiers	23
Contextual Input Keys	24
12. Troubleshooting And Current Limitations	25
The 3D View Does Not Start	25
A Boolean, Fillet, Or Chamfer Fails	25

A Sketch Will Not Extrude 25

A Sweep Will Not Build 25

A Projection Is Empty 25

Import Or Export Issues 25

Unit Display 26

Shortcut Focus And Browser Conflicts 26

Quick Recovery Checklist 26

1. Getting Started

What SketchForge Does

SketchForge is a browser-based modeler for creating parts from primitives or custom 2D profiles. It supports solid and hole workflows, boolean grouping, construction and placement planes, fillets and chamfers, sketch extrusion, revolve, sweep, and common fabrication formats.

Projects normally stay on the device in browser storage. No account is required. A server deployment can optionally provide shared SKF project storage and controlled local-folder exports.

Browser Requirements

- A current browser with WebGL 2, IndexedDB, Web Workers, and WebAssembly.
- Hardware acceleration enabled for the 3D viewport and revolve preview.
- Enough local storage for imported files and project history.
- Network access is not required after the application and its local worker assets have loaded.

Create Your First Model

1. Open the dashboard and select **Create new 3D design**.
2. In the editor, stay in the **Geometry** workspace.
3. Open **Shapes** and choose **Box**.
4. Select the box and use the handles or Shape Inspector to change its size and position.
5. Add a cylinder, change it to a **Hole**, and move it through the box.
6. Select both objects and choose **Group** to subtract the hole.
7. Open **Export** and choose STL or 3MF for 3D printing.

Dashboard Basics

- **Create:** starts a new local project.
- **Open:** opens a local project card or an SKF package.
- **Import geometry:** creates a project from STL, 3MF, STEP/STP, or SVG files.
- **Rename:** changes the local project name.
- **Delete:** permanently removes the selected local project after confirmation.
- **Search and sort:** filter by name and sort by recent activity or name.
- **Grid/List:** change the project-card layout.
- **Shared:** available only when shared project storage is configured by the server administrator.

Projects autosave geometry, history, imported assets, workspace settings, and thumbnails. Opening a shared SKF file creates a local editable copy. Saving back to shared storage checks the server revision before replacing the file.

2. Navigating The Editor

Main Areas

- **Top toolbar:** project, clipboard, history, shape, visibility, combine, modify, arrange, and output commands.
- **3D viewport:** selection, camera navigation, object transforms, workplane placement, and ruler tools.
- **Shape Inspector:** object type, dimensions, position, rotation, color, solid/hole state, sketch settings, and special properties.
- **Workspace tabs:** switch between Geometry and Sketch workflows.
- **Snap Grid:** sets movement and sketch precision.
- **Status message:** reports completed actions, requirements, and errors.

Mouse And Camera Controls

Input	Action
Left click	Select a shape or use an active handle/tool.
Shift + left click	Add or remove a shape from the selection.
Drag empty viewport	Marquee-select intersecting shapes.
Shift + marquee	Add marquee hits to the existing selection.
Right drag	Orbit the 3D camera.
Middle drag	Pan the camera.
Ctrl/Command + left drag	Pan the camera.
Mouse wheel	Zoom toward the pointer.
View cube	Switch to a face, edge, or corner view.

Use **Home**, **F**, or the **Home** key to reset the camera. Press **O** to toggle perspective and orthographic projection. Use **+** and **-** to zoom without the mouse wheel.

Camera Home restores a fixed default view. It is not a fit-selection command.

3. Creating And Editing 3D Geometry

Primitive Shape Library

Tool	Description	Useful properties
Box	Rectangular solid, cutter, or rounded box.	Width, depth, height, corner radius, steps.
Cylinder	Circular or elliptical prism.	Width, depth, height, sides.
Sphere	Ellipsoid generated from a sphere.	Width, depth, height, resolution steps.
Cone	Cone or frustum.	Base radius, top radius, depth scale, sides.
Pyramid	Pointed polygonal solid.	Width, depth, height, sides.
Wedge	Sloped block.	Width, depth, height.
Text	Extruded 3D text.	Text, font, bevel, segments, dimensions.
Round Roof	Half-round roof profile.	Dimensions and side resolution.
Half Sphere	Upper half of an ellipsoid.	Dimensions and resolution.
Torus	Ring-shaped solid.	Width, depth, height.
Tube	Hollow cylindrical body.	Dimensions and wall thickness.
Gear	Spur, helical, or bevel-style gear.	Teeth, tooth size/width, center hole, helix angle and quality.

New shapes are inserted on the active placement workplane. Their initial orientation follows that plane.

Selection

- Click a shape to replace the selection.
- Shift-click to toggle another shape.
- Drag on empty space to create a selection marquee.
- Press **Escape** or click empty space to clear the selection.
- Press **Mod+A** to select all visible shapes.
- Hidden shapes are not selected by Select All.
- Locked shapes can be selected but cannot be moved through normal transform handles.

Transform Handles

Selected shapes can expose controls for:

- Moving along the active workplane axes.
- Changing width and depth.
- Changing height.
- Raising or lowering elevation.
- Rotating around X, Y, or Z.
- Entering exact dimensions, movement distances, and angles.

Hold **Shift** while dragging a corner resize handle to preserve width/depth proportions. Hold **Alt/Option** to resize around the center. Hold both modifiers to combine those behaviors. Hold **Shift** while rotating to snap to 45-degree increments.

Shape Inspector

The inspector shows only the controls supported by the current object. Common controls include:

- Name, color, visibility, and lock state.
- Solid or Hole mode.
- Width, depth/length, height, elevation, and position.
- X, Y, and Z rotation.
- Snap precision.
- Shape-specific radius, side, resolution, wall, text, or gear settings.

Press **Enter** after typing a numeric property to commit it. Press **Escape** to restore the prior displayed value.

Moving With The Keyboard

- Arrow keys move selected unlocked shapes on the active placement plane.
- Hold Shift for 5-unit steps instead of 1-unit steps.
- Hold Mod with Up/Down to move along the placement-plane normal.
- Press **D** to drop selected unlocked shapes onto the active placement workplane.

Clipboard And History

- Copy, cut, paste, duplicate, and delete work on selected 3D shapes.
- Paste creates new IDs and offsets the copies so they are visible.
- Clipboard data can be shared between SketchForge tabs when browser permissions allow.
- Undo and redo are separate from the temporary history used while editing a sketch.
- History retention is configurable in Workspace Settings.

Delete and Cut can remove selected locked objects. Lock protects normal transforms and inspector edits, not every destructive command.

Visibility, Locking, And Separation

- **Hide selected:** hides unlocked selected shapes.
- **Show selected:** restores selected hidden shapes.
- **Show all hidden:** restores every hidden shape.
- **Lock:** prevents normal transforms and property changes.
- **Separate:** splits a mesh or grouped object into disconnected components when multiple components exist.

4. Placement And Construction Planes

SketchForge uses two related plane systems.

Placement Workplane

The placement workplane controls where new objects are inserted and how movement, nudging, and Drop to Workplane are oriented.

1. Select **Place workplane** or press **W**.
2. Click a planar model face.
3. Hold Shift while clicking to reverse the face normal.
4. Click empty grid space to return the placement workplane to the base horizontal plane.

Press **Shift+W** with exactly one visible shape selected to place the workplane on its local top face. Press **Escape** to cancel placement mode.

Picking a model face also creates and activates an associative construction plane for sketching.

Persistent Construction Planes

Open **Geometry > Workplane** to manage sketch planes:

- Base XZ plane.
- Offset XZ/top plane.
- Offset XY/front plane.
- Offset YZ/side plane.
- Planes attached to planar model faces.

Face-attached construction planes follow movement, rotation, and resizing of the source object. Construction-plane helpers are locked and omitted from normal geometry exports.

Plane Workflow Tips

- Select the intended construction plane before starting a sketch.
- Use a persistent construction plane for angled projection, sweep, or revolve workflows.
- Use placement planes for quick object insertion and movement orientation.
- Reset the persistent sketch plane from the Construction Planes panel when returning to Base XZ.

5. Combining And Modifying Geometry

Solids And Holes

Any normal shape can act as a cutter:

1. Select the cutter shape.
2. Choose **Hole** or press **H**.
3. Position it so it intersects a solid.
4. Select the solid and hole together.
5. Choose **Group**.

Press **S** to change selected unlocked holes back to solids. Standalone holes are translucent and omitted from normal mesh/vector exports.

Group And Ungroup

- Group requires at least two selected unlocked objects.
- A solid-only group creates one grouped object.
- A group containing solids and holes performs subtraction.
- Ungroup restores stored operands when available.
- If a hole completely consumes its solid, the operation can remove all selected operands. Use Undo to recover them.

Boolean Intersection

Intersection requires at least one unlocked solid and one unlocked hole. The result keeps their overlapping volume.

If the intersection is empty, the selected operands may be removed. Use Undo if this was not intended.

Align

Align requires at least two selected shapes.

1. Enter Align mode or press **L**.
2. Choose minimum, center, or maximum alignment on X, Y, or Z.
3. Hover an alignment point to preview.
4. Click an object to make it the anchor where available.

A locked selected object acts as the fixed anchor. Align mode and Mirror mode are mutually exclusive.

Mirror

Mirror transforms selected unlocked shapes in place around the combined selection center.

1. Select one or more shapes.
2. Enter Mirror mode or press **M**.
3. Choose X, Y, or Z.
4. Review the preview and apply.

Mirror does not create copies. Duplicate first when both versions are required.

Fillet And Chamfer

Requirements:

- Exactly one selected object.
- The object is unlocked and is not a hole.
- Geometry can be prepared by the local OpenCascade worker.

Workflow:

1. Choose **Fillet** or **Chamfer**.
2. Wait for edge preparation.
3. Click highlighted edges.
4. Hold Shift to toggle only the clicked edge rather than its tangent chain.
5. Adjust radius/distance, chamfer angle, sharp-edge threshold, tangent-chain behavior, resize behavior, and preview quality.
6. Select **Apply** or press **Enter**. Press **Escape** to cancel.

Applied features retain removable history when possible. Removing an older feature also removes newer features that depend on it.

Boolean And Edge Limitations

Operations can fail on open, non-manifold, very complex, or unsuitable imported meshes. SketchForge attempts exact, Manifold, and fallback methods, but not every triangle soup can become a valid solid. Edge preparation and complex STEP operations may take longer the first time because the local CAD kernel must load.

6. Measurement Tools

Selection Dimensions

Transform overlays can show editable width, depth, height, elevation, rotation, and optional movement distances. Press Enter to commit a direct numeric edit and Escape to cancel it.

When movement dimensions are enabled, use Tab and Shift+Tab to move between settled movement fields.

3D Ruler

Open **Ruler tools** and choose:

- Add measurement.
- Move measurement points.
- Delete measurement parts.

Ruler points can attach to vertices, edges, and model surfaces. Clicking an edge can measure its full polyline length. Attached measurements follow transforms when the referenced topology remains available.

Ruler annotations are viewport-session state, not project geometry.

Sketch Measure And Dimension

- **Measure:** temporary two-point distance. It does not constrain geometry.
- **Dimension:** persistent driving or reference dimensions stored with the sketch.

The underlying model uses millimeters. Some inspector controls convert display units, while sketch dimensions, ruler values, and transform overlays currently show raw model/mm values.

7. The 2D Sketch Workspace

Start Or Edit A Sketch

1. Switch to **Sketch**.
2. Open **Sketch to 3D**.
3. Choose **Extrude sketch** or **Revolve sketch**.
4. Draw on the active construction plane.
5. Finish to create the 3D object, or cancel to discard the sketch.

To edit a sketch-derived object, select exactly one such object and choose **Edit sketch**. Finishing replaces the same 3D object. Canceling discards the edit.

Sketches are stored on their generated 3D objects rather than as separate dashboard documents.

Sketch Navigation

Input	Action
Mouse wheel	Zoom the sketch plate.
Middle drag	Pan.
Ctrl/Command + left drag	Pan.
Escape	Clear the active draft, chain, and selection.
Delete/Backspace	Delete selected sketch entities.
Mod+Z	Sketch undo.
Mod+Shift+Z or Mod+Y	Sketch redo.

Escape does not exit the entire sketch. Use the Finish or Cancel controls for that.

Drawing Tools

Tool	How to use it	Notes
Select	Click or marquee geometry, points, images, text, and regions.	Linked projection geometry is fixed.
Line	Click successive points; click an existing endpoint to connect or close.	Closing a loop switches to Select.
Bezier Curve	Click-drag points to pull symmetric handles.	Handles can be edited later.
Smooth Curve	Click points along a flowing path.	Handles are regenerated for smooth tangency.
Center Circle	Click center, then radius point.	Stored as four editable Bezier segments.
Diameter Circle	Click opposite diameter endpoints.	Stored as four editable Bezier segments.
Corner Rectangle	Click opposite corners.	Horizontal/vertical constraints are added.
Center Rectangle	Click center, then a corner.	Horizontal/vertical constraints are added.
Polygon	Set 3-24 sides, click center, then a vertex.	Creates a regular closed profile.
Text	Click, type, then press Enter or click away.	Converts text to closed line contours.
Refine	Click a segment to add a point; click a point to remove it.	Two neighbors are reconnected where possible.
Erase	Click a point, segment, or visible dimension to remove it.	Use Delete for selected entities.
Dimension	Create driving line lengths or two-anchor references.	Persistent with the sketch.
Measure	Pick two positions for a temporary distance.	Does not constrain geometry.

Point And Curve Editing

Select a point to change its mode:

- **Corner:** removes handles and uses straight incident segments.
- **Smooth:** creates linked opposing curve handles.
- **Split:** allows incoming and outgoing handles to move independently.
- **Fix:** locks the point at its current coordinates.

Snapping

Sketch snapping can use:

- Existing endpoints.
- Segment midpoints.
- Closed-profile centers.
- Text anchors.
- X/Z alignment guides through candidates.
- The configured Snap Grid.

- Optional visible grid-line magnetism.

Exact geometry anchors take priority. Endpoint snaps can reuse existing points and create connected topology. Midpoint and alignment snaps position geometry but do not create persistent constraints.

Constraints

Select a straight segment to apply Horizontal or Vertical. Applying one removes the opposite constraint. Rectangles receive both automatically where appropriate.

Select a point and use Fix to hold it in place. Moving constrained geometry invokes the solver and may move connected points. Current sketch constraints do not include tangent, parallel, perpendicular, equal, angle, radius, or diameter.

Dimensions

Driving Line Length

1. Choose Dimension.
2. Click a straight line.
3. Enter a positive Length.
4. Set, update, or remove the dimension.

The solver adjusts endpoint geometry. In Select mode, the dimension label can be repositioned.

Reference Distance

With Dimension active, select two anchors from endpoints, midpoints, or intersections. Reference dimensions update with geometry but do not drive it. Duplicate and zero-length references are rejected.

Reference Images

1. Use Select and choose **Add Image**.
2. Choose PNG, JPEG, WebP, GIF, BMP, or SVG.
3. Drag the image or use its handles.
4. Adjust width, height, opacity, position, and aspect lock in the inspector.

Images are references only. They do not create extrusion regions.

Sketch Text

Sketch text converts immediately to geometric line contours. It can form regions and can be refined, erased, mirrored, or patterned. There is no semantic text/font/size editor after placement.

Project Geometry Into A Sketch

1. Choose **Project** in an active sketch.
2. Select a source sketch-derived object or 3D shape.
3. Choose **Linked reference** or **Editable copy**.
4. Apply Project.

A sketch source projects its native profile. A 3D shape creates a plane intersection, not a silhouette. Linked references refresh when the sketch reopens and cannot be directly edited.

Offset

Select one or more non-branching connected segments, choose Offset, enter a nonzero distance, and optionally include the connected path. Positive values offset closed loops outward and open paths to their left. Bezier curves are flattened into line segments in the result.

Sketch Mirror And Patterns

- **Mirror:** copy around Origin X, Origin Z, or a selected segment.
- **Rectangular pattern:** 1-20 rows/columns with configurable spacing and optional selected-line direction.
- **Circular pattern:** 2-40 instances around the sketch origin or a selected point, over a nonzero total angle.

These operations keep the originals and select the generated copies. Compatible constraints and dimensions are remapped where possible.

8. Turning Sketches Into 3D Models

Extrude

1. Start Extrude Sketch.
2. Draw one or more closed regions.
3. In Select mode, click shaded regions or use All/Clear.
4. Finish the sketch.
5. Adjust Height on the resulting shape.

Regions support disjoint solids, nested holes, islands, overlaps, and open dividers crossing a closed face. At least one valid region must be selected. New extrusions start at 10 mm and extend along the positive plane normal.

SketchForge first requests exact OpenCascade/B-Rep geometry. If that fails, it falls back to a triangulated mesh that remains sketch-editable but may not retain exact CAD edges.

Revolve

1. Start Revolve Sketch.
2. Draw on the left side of the marked vertical revolve axis.
3. Review the live 3D preview.
4. Finish the revolve.
5. Adjust start angle, sweep angle, side count, and open-path thickness.

Closed paths create filled sections. Open paths use the Thickness setting. Geometry crossing the axis is clipped and geometry fully on the unusable side is ignored. Revolve creates Manifold mesh geometry rather than an exact OCCT B-Rep.

Sweep

1. Draw exactly one closed section and one separate open path.
2. Use Select and marquee every segment in both paths.
3. Choose Sweep.
4. Confirm Create Sweep.

Sweep supports straight and Bezier paths. It creates the body immediately and stores section/path IDs for later sketch editing. Holes, multiple sections, multiple paths, branching, and connected section/path topology are not supported.

9. Import, Export, And Project Files

Import Formats

Format	Support	Notes
STL	Import	Triangulated mesh.
3MF	Import	Package transforms, declared units, mesh geometry, and display color.
STEP/STP	Import	Requires solid geometry and retains B-Rep data.
SVG	Import	Converts valid closed vector geometry to a shallow extrusion.
SKF	Open project	Restores editable project state and assets.
OBJ	Not imported	OBJ is export-only.

Multiple geometry files can be imported together. Importing from the dashboard creates a new local project.

Export Formats

Format	Best use	Behavior
STL	Slicers and printing	Triangulated mesh.
3MF	Modern print workflow	Millimeter package with object names and colors.
OBJ	Mesh interchange	General triangulated geometry.
SVG	2D manufacturing	Top-view silhouette with holes.
STEP	CAD exchange	Exact supported primitives and retained STEP B-Reps.
SKF	Editable backup/share	Geometry, sketches, history, assets, settings, and CAD data.

If shapes are selected, direct exports use selected solids. With no selection they use all non-hole, non-construction-plane shapes. Hidden solids can be included in an all-project export, so select the intended objects when hidden content must be excluded.

Exact STEP export is currently limited to supported boxes, cylinders, spheres, cones, and imported STEP bodies that retained B-Rep data. Unsupported mesh and swept geometry is reported as skipped.

SKF Project Packages

SKF can preserve:

- Native and imported geometry.
- Groups and boolean operands.
- Sketch profiles, constraints, dimensions, projections, images, and feature metadata.
- Construction and placement planes.
- Undo/redo history according to the chosen limit.
- Workspace settings and themes.

- Original imported source assets and exact CAD data.

Opening an SKF package creates a new local project rather than overwriting the currently open one.

Download Destinations

- **Browser downloads:** available in normal and static deployments.
- **Local folder:** available in localhost server builds. The folder must already exist and be inside Downloads or **SKETCHFORGE_LOCAL_DOWNLOAD_ROOT**.
- **Shared storage:** SKF-only and available when configured by the server administrator.

10. Workspace Settings

Appearance

- SketchForge, Light, Dark, SolidWorks, Inventor, and Custom themes.
- Custom UI, background, grid, shape, selection, hole, and outline colors.
- Shadows.
- Movement dimensions.
- Camera zoom speed.
- Cruise while adding shapes setting.

Measurement

- Metric, Imperial, or Bricks.
- Millimeters, centimeters, meters, inches, feet, or studs.
- One to three decimal places.
- Snap Grid from Off through fine millimeter steps and Brick/8 mm.

Workplane

- Width and depth from 60 to 2000 mm.
- Grid block size from 1 to 200 mm.
- Preset or custom grid color.

History

- 30, 50, 100, Unlimited, or a custom limit up to 5000 actions.
- Lowering the limit permanently removes older history states from the current project.

Dashboard Export Settings

Dashboard Settings chooses between normal browser downloads and an approved local-folder destination. It does not change local project autosave.

11. Complete Keyboard Shortcut Reference

Modifier Names

In the tables below, **Mod** means:

- macOS: Command
- Windows/Linux: Ctrl

Application shortcuts are ignored while typing in an input, text area, select, or content-editable field. Extra modifiers are accepted by several shortcuts, so prefer the exact combinations shown.

Project And Selection Shortcuts

Shortcut	Context	Action
Mod+Z	Geometry or Sketch	Undo in the active mode.
Mod+Shift+Z	Geometry or Sketch	Redo in the active mode.
Mod+Y	Geometry or Sketch	Alternate redo.
Mod+C	Geometry	Copy selected shapes.
Mod+X	Geometry	Cut selected shapes.
Mod+V	Geometry	Paste from available clipboard data.
Mod+D	Geometry	Duplicate selected shapes.
Mod+A	Geometry	Select all visible shapes.
Mod+G	Geometry	Group selected unlocked shapes.
Mod+Shift+G	Geometry	Ungroup selected groups.
Mod+L	Geometry	Toggle lock for the selection.
Mod+H	Geometry	Toggle selected visibility.
Mod+Shift+H	Geometry	Show all hidden shapes.
Delete or Backspace	Geometry	Delete selected shapes.
Delete or Backspace	Sketch	Delete selected sketch entities; otherwise clear a temporary measurement.
Escape	Geometry	Clear shape selection; active contextual tools may also cancel.
Escape	Sketch	Clear the active draft, chain, and selection without exiting the sketch.

Movement And Modeling Shortcuts

Shortcut	Context	Action
Left / Right	Geometry	Nudge along placement-plane X by 1.
Up / Down	Geometry	Nudge along placement-plane Z by 1.
Shift+Arrow	Geometry	Nudge by 5 instead of 1.
Mod+Up / Mod+Down	Geometry	Move by 1 along the placement-plane normal.
Mod+Shift+Up / Mod+Shift+Down	Geometry	Move by 5 along the placement-plane normal.
D	Geometry	Drop selected unlocked shapes to the placement workplane.
H	Geometry	Change selected unlocked shapes to holes.
S	Geometry	Change selected unlocked shapes to solids.
L	Geometry	Toggle Align mode.
M	Geometry	Toggle Mirror mode.

S is Make Solid, not Save. Projects autosave and SketchForge has no manual Save shortcut. If a browser/operating-system combination such as Mod+S or Mod+M reaches the editor, the matching modeling command can run.

Viewport And Workplane Shortcuts

Shortcut	Context	Action
W	3D viewport	Toggle placement-workplane mode.
Shift+W	3D viewport	Place on the selected shape's top face when exactly one visible shape is selected; otherwise toggle placement mode.
F	3D viewport	Reset camera to Home.
Home	3D viewport	Reset camera to Home.
O	3D viewport	Toggle perspective/orthographic projection.
= or +	3D viewport	Zoom in.
- or _	3D viewport	Zoom out.
Escape	3D viewport	Cancel workplane placement or reset active ruler modes.

Pointer Modifiers

Input	Context	Action
Mod + left drag	3D viewport or Sketch	Pan.
Middle drag	3D viewport or Sketch	Pan.
Right drag	3D viewport	Orbit.
Shift + shape click	3D viewport	Toggle shape in selection.
Shift + empty marquee	3D viewport	Add marquee hits to selection.
Shift + edge click	Fillet/Chamfer	Toggle only one edge instead of its tangent chain.
Shift + face click	Workplane placement	Reverse the picked face normal.
Shift + corner resize	Transform	Preserve width/depth proportions.
Alt/Option + resize	Transform	Resize around the center.
Shift + Alt/Option + corner resize	Transform	Proportional center-based resize.
Shift + rotate drag	Transform	Snap rotation to 45-degree increments.

Transform modifiers can be pressed or released during the drag.

Contextual Input Keys

Key	Context	Action
Enter	Transform, movement, inspector, edge field	Commit the typed value, usually by leaving the field.
Escape	Transform or edge numeric field	Cancel or restore the prior value.
Tab / Shift+Tab	Movement dimensions	Move forward/backward between available movement fields.
Enter	Sketch text	Commit non-empty text.
Escape	Sketch text	Cancel text placement.
Enter or Space	Focused sketch region	Toggle extrusion-region selection.
Enter	Sketch driving-length field	Apply the typed straight-line length.
Enter	Sketch image property	Commit width, height, opacity, or position.
Enter	Sweep, Project, Offset, Mirror, or Pattern form	Submit the active sketch operation.
Enter	Export filename	Run the selected export when allowed.
Enter	Active Fillet/Chamfer session	Apply a valid prepared preview.
Escape	Active Fillet/Chamfer session	Cancel the modifier session.
Enter	Workspace numeric setting	Commit width, length, grid size, or custom history limit.
Escape	Grid-color popover	Close the color popover and return focus.
Enter	Dashboard rename field	Confirm the project rename.
Escape	Sketch-create or Visibility menu	Close the open toolbar menu.

Most full dialogs do not use Escape as a universal close command. Use their visible Close or Cancel control. One Escape keypress can clear selection while also canceling an active workplane, ruler, edge, or toolbar state.

12. Troubleshooting And Current Limitations

The 3D View Does Not Start

- Confirm hardware acceleration and WebGL 2 are enabled.
- Close other graphics-heavy tabs or applications.
- Use the on-screen retry control.
- Update the browser and graphics driver.
- Revolve preview can show a fallback when WebGL is unavailable.

A Boolean, Fillet, Or Chamfer Fails

- Confirm the object is a closed solid rather than an open mesh.
- Simplify highly detailed imported geometry.
- Check that a hole actually intersects its solid.
- For edge tools, select exactly one unlocked solid.
- Try Draft preview quality before Standard or Fine.
- Undo if an empty boolean removed the operands.

A Sketch Will Not Extrude

- Confirm at least one shaded bounded region exists and is selected.
- Close small endpoint gaps.
- Remove zero-length or duplicate segments.
- Avoid self-intersecting loops unless the intended split regions are visible.
- Open-only geometry cannot form a normal extrusion region.

A Sweep Will Not Build

- Select exactly one closed section and one separate open path.
- Include every segment in both paths.
- Remove branches and shared endpoints between section and path.
- Holes and multiple sweep paths are not currently supported.

A Projection Is Empty

- A 3D source must intersect the active sketch plane.
- Projection is a plane intersection, not a camera silhouette.
- Geometry perpendicular to the plane can collapse.
- Prefer a persistent construction plane for angled workflows.

Import Or Export Issues

- OBJ is export-only.

- STEP export skips unsupported mesh/text/sweep bodies rather than faceting them.
- 3MF and SKF enforce archive and asset safety limits.
- Local-folder export works only in an eligible localhost server deployment and approved directory.
- Select intended visible objects before export when hidden objects must not be included.

Unit Display

The model kernel uses millimeters. Inspector and edge-modifier fields support configured display units, but sketch dimensions, 3D ruler values, transform labels, and movement overlays currently display raw model/mm values.

Shortcut Focus And Browser Conflicts

Shortcuts are suppressed while typing in standard fields. They can still act when a toolbar button has focus. Several browser combinations share letters with SketchForge commands. Use the exact shortcuts in this manual and click the viewport before issuing model commands.

Quick Recovery Checklist

1. Read the status message at the bottom of the editor.
2. Press Escape to cancel the active contextual tool.
3. Check selection count, lock state, solid/hole mode, and active plane.
4. Use Undo before trying a different operation.
5. Save an SKF package before complex booleans or imported-mesh operations.

SketchForge is an open-source project. For installation, server deployment, and contributor information, see the repository README.